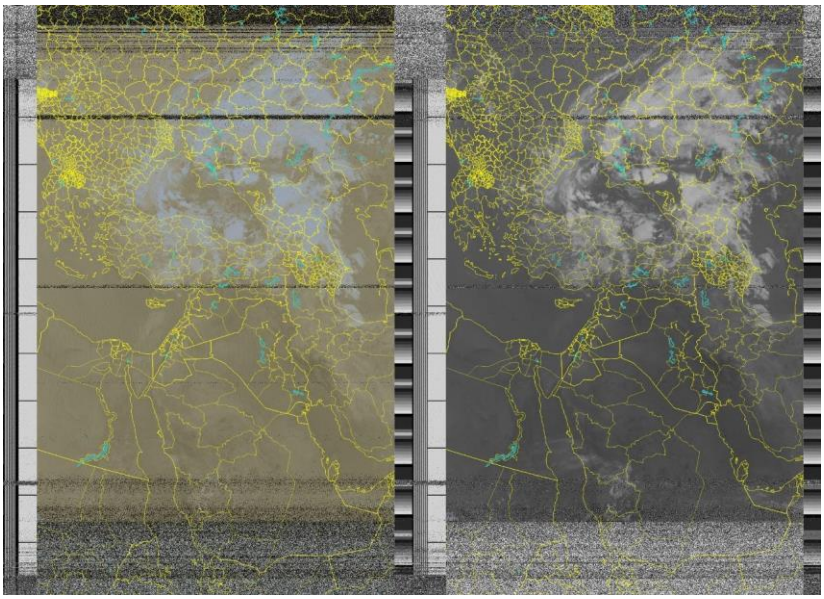
Previous Projects

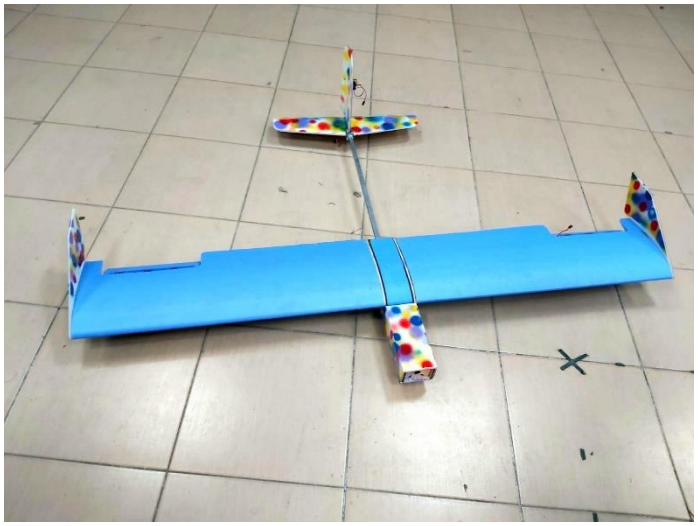
(1) Design, Fabrication and Testing of LEO Ground Station [2023]



It was the 3rd Year Project, it consists of three main parts: propagator, communication subsystem and rotator, we built our own propagator based on SGP4 model, designed and manufactured a Yagi Antenna and built a rotator with two DOFs and implemented a controller for it and finally were able to download an image of Egypt from the weather satellite NOAA-18.

Full Documentation of the project: Drive Link

(2) A UAV capable of towing a banner with the minimum possible [2021]



I was responsible for the matching plot, matching plot and the release and deployment mechanism of the banner. I also participated in the manufacturing process which included laser cutting for the fuselage and flat parts and hot wire machine for fabricating the wing and the tail. We ranked third in the competition held by Unmanned Aerial Systems Development Center (UDC) at Cairo University.

Documentation of the project: Drive Link

(3) CanSat Design and Fabrication [2021]

I was responsible for the control and data measurement. Arduino was used as a MCU and I wrote libraries for the following sensors: DHT11, BMP180, Ublox NEO-6M GPS and MPU6050. I also participated with my colleagues in the manufacturing process of PCBs and the satellite bus. This was a part of a training and a competition held by Space Systems Technology Laboratory (SSTL) at Cairo University.